

Onboarding Exercises - SURP 2026

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Introduction

These exercises are designed to build the necessary galpy skills in order to simulate the dynamical perturbation of the Milky Way halo by the Large Magellanic Cloud (LMC).

1 Exercise 1

1.1

Exercise 1.1 is to plot the galpy potential MWPotential2014's circular-velocity curve from 0.1-100kpc, as well as its bulge, disk, and halo contributions, with the results shown in Figure 1.1. I used the plotRotCurve method to plot each potential. I also marked the circular velocity at $R_0 = 8\text{kpc}$.

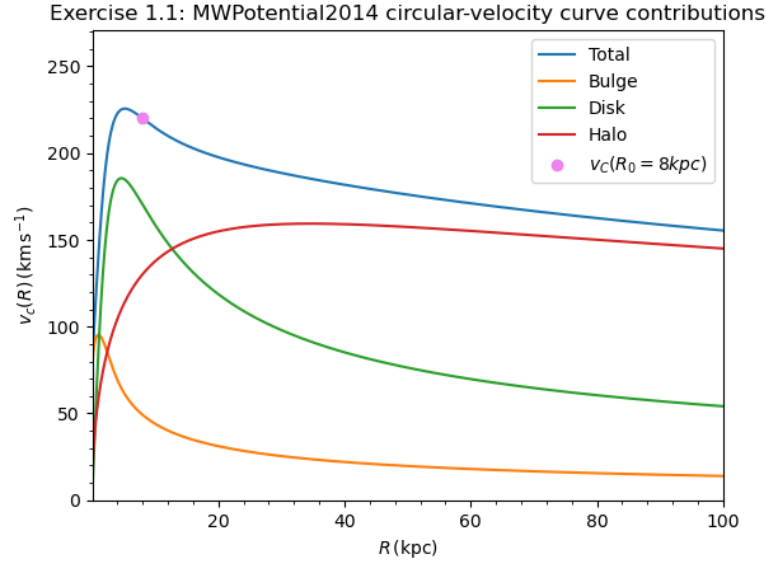


Figure 1.1: Exercise 1, question 1: MWPotential2014 circular velocity curve with bulge, disk, and halo components

1.2

Exercise 1.2 is to plot $\rho(r)$ along the disk's major axis ($z=0$), shown in Figure 1.2, as well as $M_{enc}(< r)$, shown in Figure 1.3, for MWPotential2014. I created arrays for each plot along a logspace grid of r values ranging from 0.1-100kpc, evaluating the density and enclosed mass respectively at each r value.

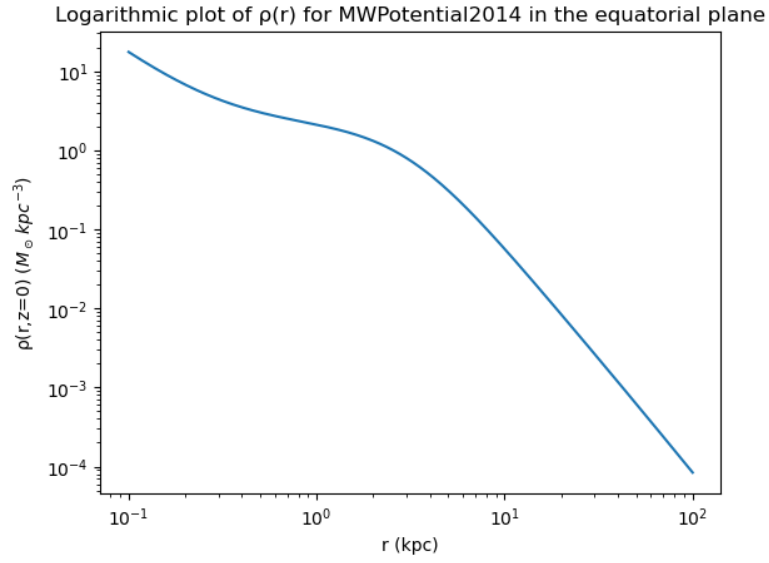


Figure 1.2: Exercise 1, question 2: $\rho(r)$ in the equatorial plane

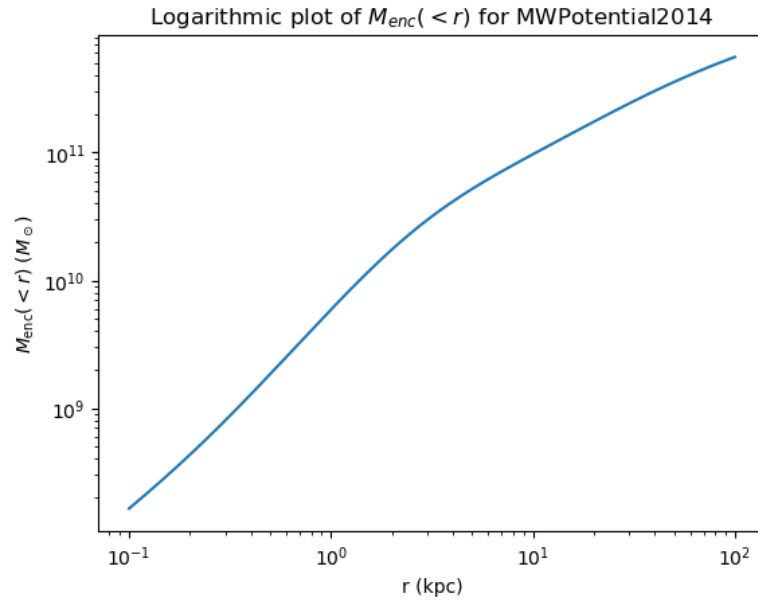


Figure 1.3: Exercise 1, question 2: $M_{enc}(< r)$

1.3

Exercise 1.3 is to overplot the circular velocity of a single Hernquist potential with a total mass of $10^{12} M_{Sun}$ and a scale radius of 30kpc on the figure from Exercise 1.1, with results shown in Figure 1.4. After initializing the Hernquist potential, I plotted it using the same method from Exercise 1.1.

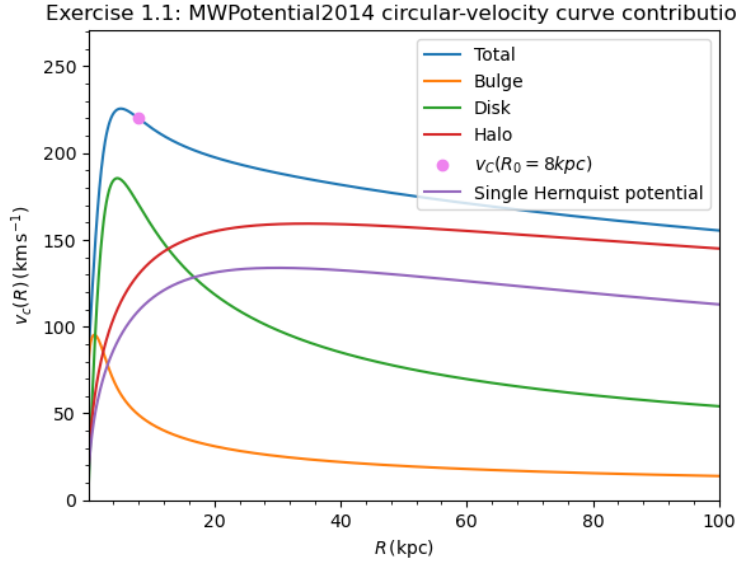


Figure 1.4: Exercise 1, question 3: Single Hernquist potential circular velocity compared with MWPotential2014 and its components

The single Hernquist potential's circular velocity curve matches the total Milky Way curve best at large R , although it is not a close match anywhere.

2 Exercise 2

2.1

Exercise 2.1's goal is to instantiate galpy's default Sun-like orbit and integrate it for 5 Gyr in MWPotential2014 with a 10,000 step time grid. To create this time grid, I used `np.linspace`.

2.2

Exercise 2.2 is to plot the orbit from 2.1 in both the (R,z) (Figure 2.1) and (x,y) (Figure 2.2) planes. Additionally, I plotted the fractional energy error for time (Figure 2.3) and z -angular momentum (Figure 2.4) with the default integrator and the original time grid. I also plotted these errors with a 1000 step time grid (Figures 2.5 and 2.6) as well as with a different integrator (Figures 2.7 and 2.8).

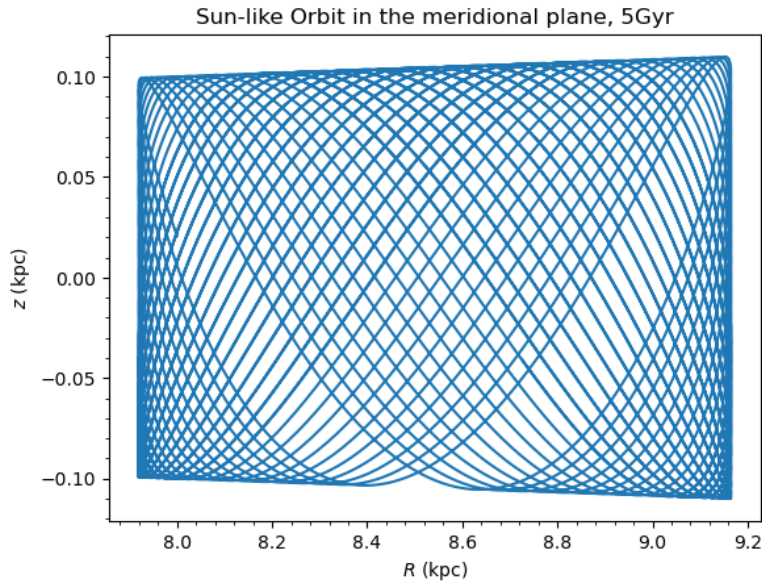


Figure 2.1: Exercise 2, question 2: Sun-like orbit in the (R,z) plane

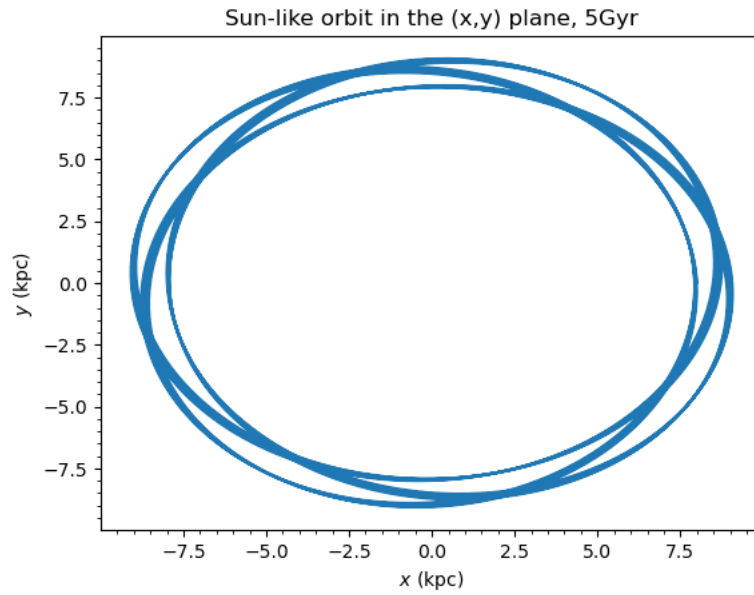


Figure 2.2: Exercise 2, question 2: Sun-like orbit in the (x,y) plane

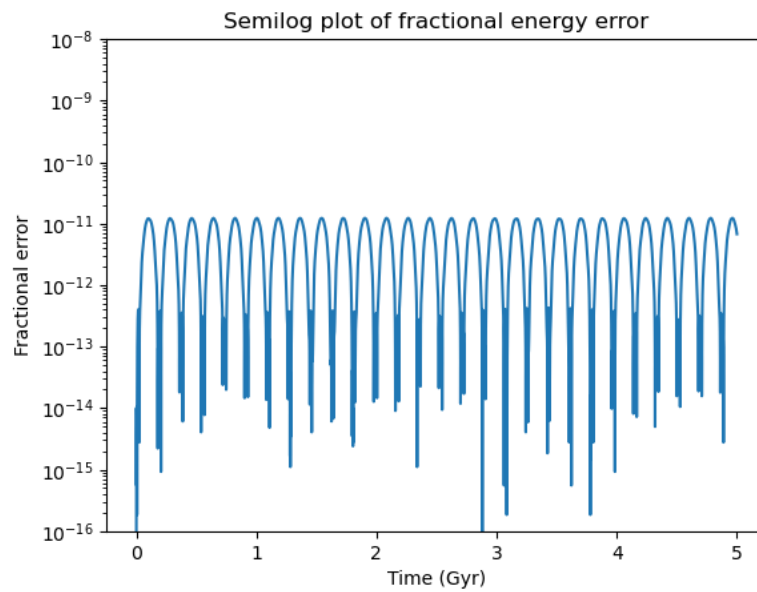


Figure 2.3: Exercise 2, question 2: Fractional energy error vs. time, 10000 step time grid, default integrator

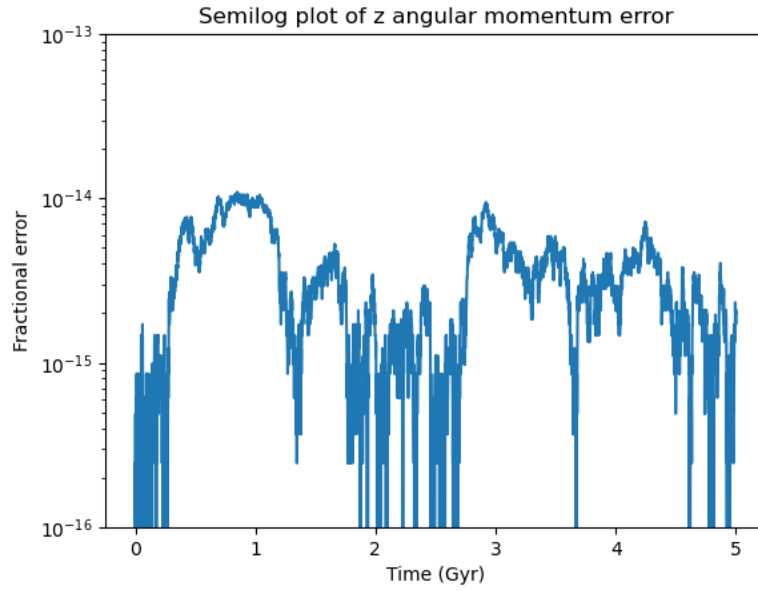


Figure 2.4: Exercise 2, question 2: Fractional z-angular-momentum error vs. time, 10000 step time grid, default integrator

Figures 2.3 and 2.4 show that the default integrator (symplec4_c) has a maximum fractional error of around 10^{-11} for energy and around 10^{-14} for z-angular-momentum (almost machine precision). These errors also appear to remain relatively stable over time.

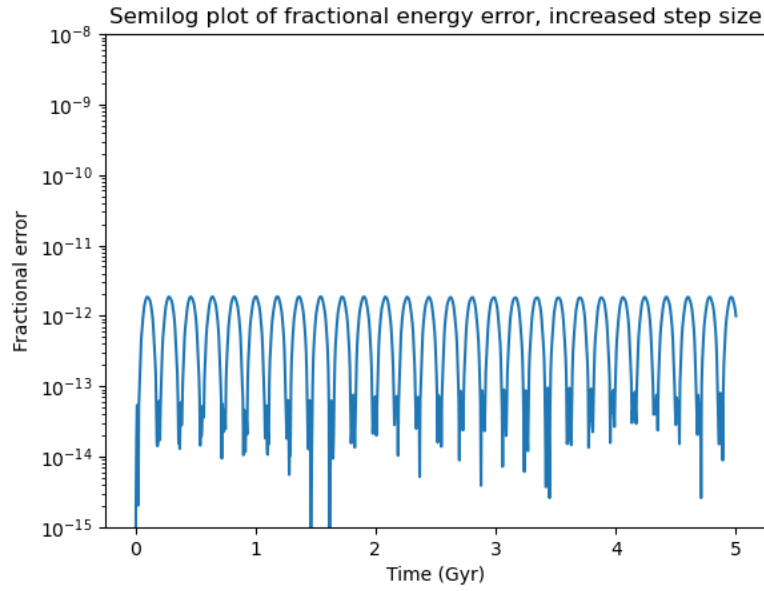


Figure 2.5: Exercise 2, question 2: Fractional energy error vs. time, 1000 step time grid, default integrator

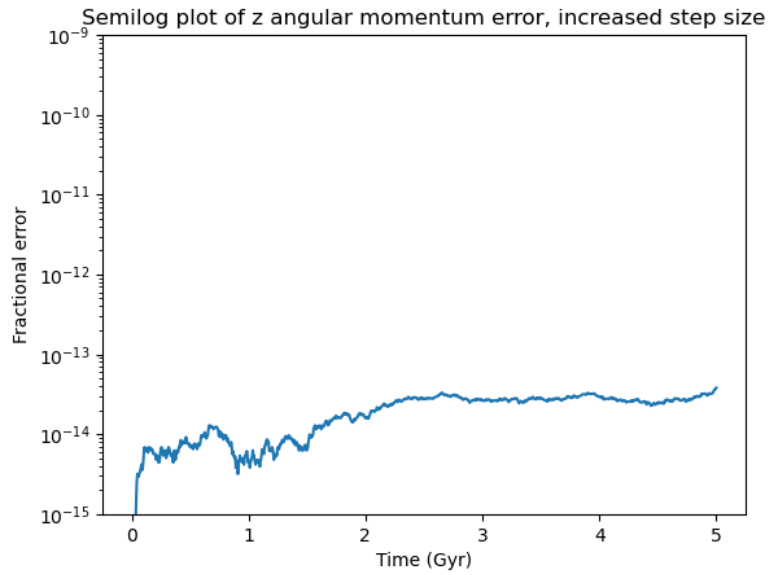


Figure 2.6: Exercise 2, question 2: Fractional z-angular-momentum error vs. time, 1000 step time grid, default integrator

Decreasing the number of steps from 10000 to 1000 did not meaningfully change the magnitude of fractional error for energy or z-angular-momentum for the default symplec4_c integrator.

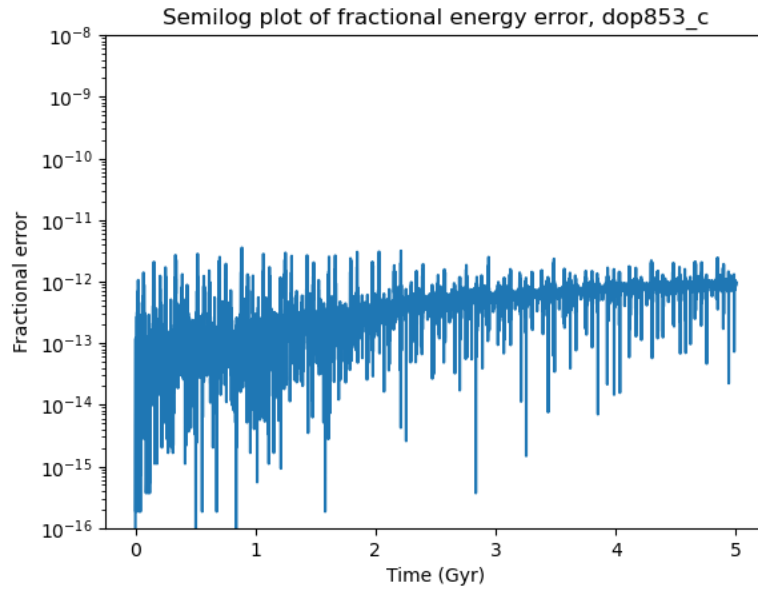


Figure 2.7: Exercise 2, question 2: Fractional energy error vs. time, 10000 step time grid, dop853_c integrator

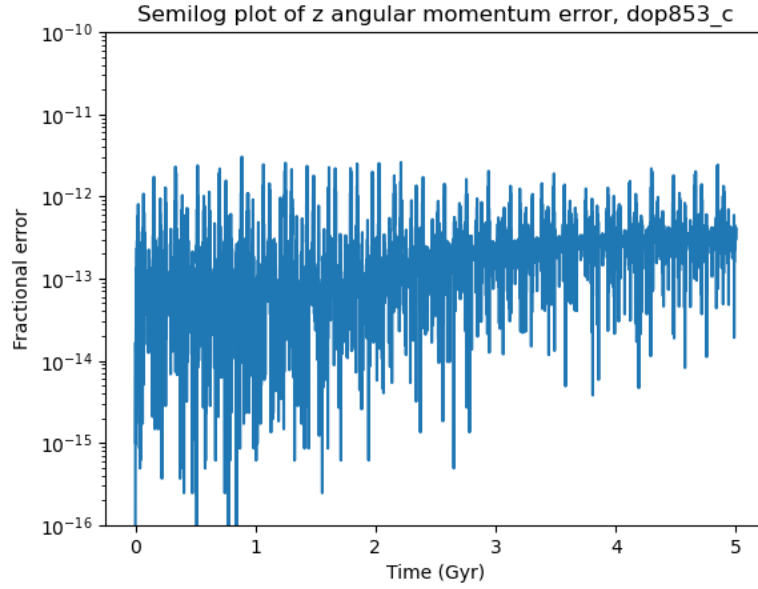


Figure 2.8: Exercise 2, question 2: Fractional z-angular-momentum error vs. time, 10000 step time grid, dop853_c integrator

The maximum fractional energy and z-angular-momentum errors for the dop853.c integrator appear slightly higher than for the default integrator.

2.3

The goal of Exercise 2.3 is to make a parameter exploration by varying initial v_T by 20% and plotting the results on the same (R,z) axes (Figure 2.9) to determine which initial conditions result in the most radial vs. the most circular orbit.

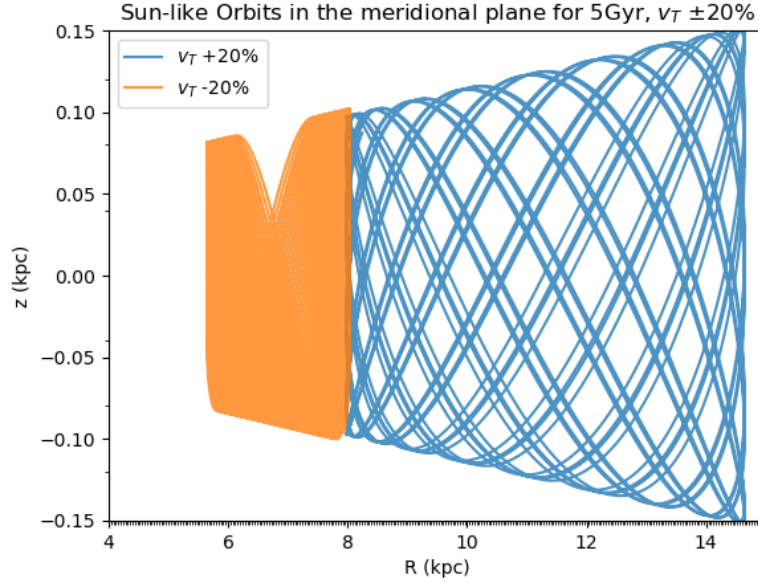


Figure 2.9: Exercise 2, question 3: v_T variation for Sun-like orbits in the meridional plane

Increasing the initial v_T by 20% produced a more radial orbit than lowering it by 20%, since Figure 2.9 shows that higher- v_T orbit has a much larger range of R values than the lower- v_T orbit.

3 Exercise 3

Exercise 3 focuses on creating a spherical Milky Way approximation whose enclosed mass closely matches a realistic Milky Way potential, with the goal of having well-defined distribution-function sampling and dynamical-friction prescriptions.

3.1

Exercise 3.1 is to compute the enclosed mass on a logarithmic grid from 0.1-300kpc, which I did using `np.logspace` and `mass(Pot,r)` from `galpy`.

3.2

The goal of Exercise 3.2 is to fit a single Hernquist potential to the enclosed mass from Exercise 3.1 by minimizing $\sum_i [\log M_{enc,Hern}(r_i) - \log M_{enc,MW}(r_i)]^2$. I did this using scipy's minimize, taking logM and loga as its parameters to keep the mass and scale radius on a similar scale for optimization.

The result for the best-fit (M, a) for the single Hernquist potential fit was $(1.0075e12 M_{sun}, 7.9348 kpc)$. The residual is shown in Figure 3.1.

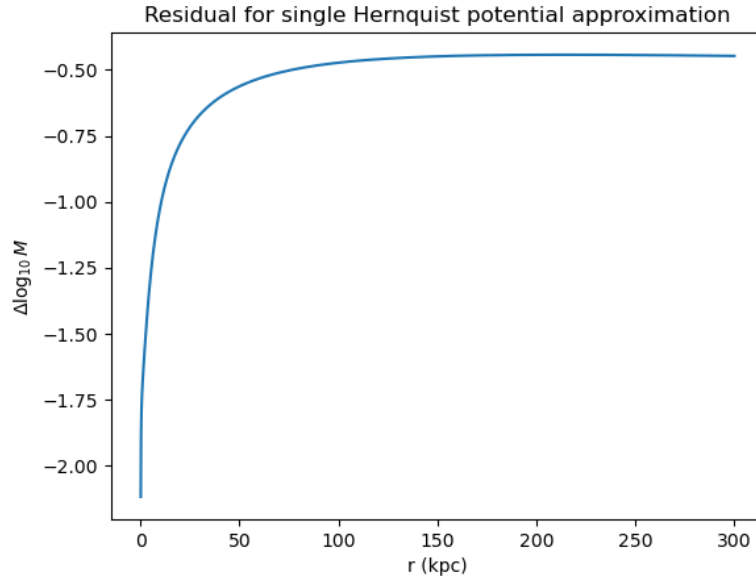


Figure 3.1: Exercise 3, question 2: Residual plot for single Hernquist approximation

3.3

Exercise 3.3 improves on Exercise 3.2 by creating a two-component proxy for the Milky Way, using Hernquist and NFW potentials. I optimized this fit similarly to the Exercise 3.2 fit, with its residual shown in Figure 3.2. As expected, it has a smaller residual than the single-potential proxy.

For the two-component fit, the best-fit (M, a) for the Hernquist (bulge) component was $(1.1676e11 M_{sun}, 2.3180 kpc)$. For the NFW (halo) component,

the best-fit (M, a) was $(3.7103e11 M_{sun}, 11.7741 kpc)$.

Additionally, I plotted $v_c(R)$ for both proxies (Figure 3.3), with the two-component proxy matching the MWPotential2014 curve more closely.

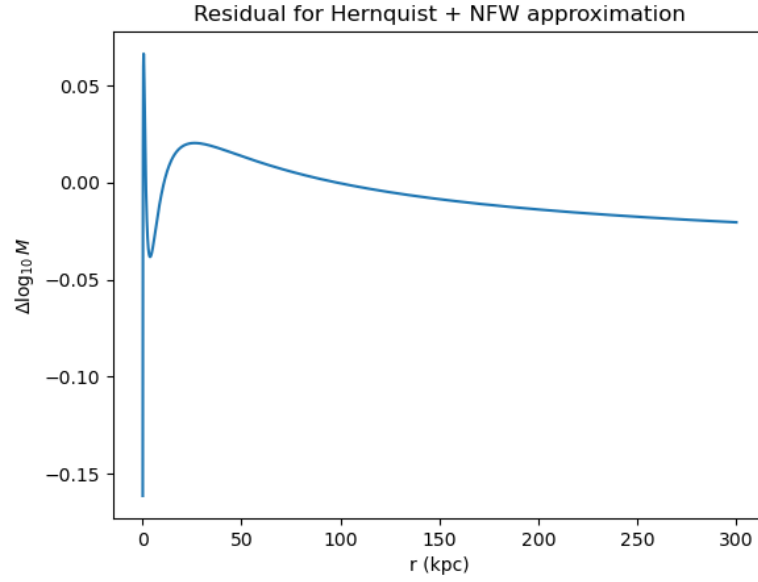


Figure 3.2: Exercise 3, question 3: Residual plot for Hernquist + NFW approximation

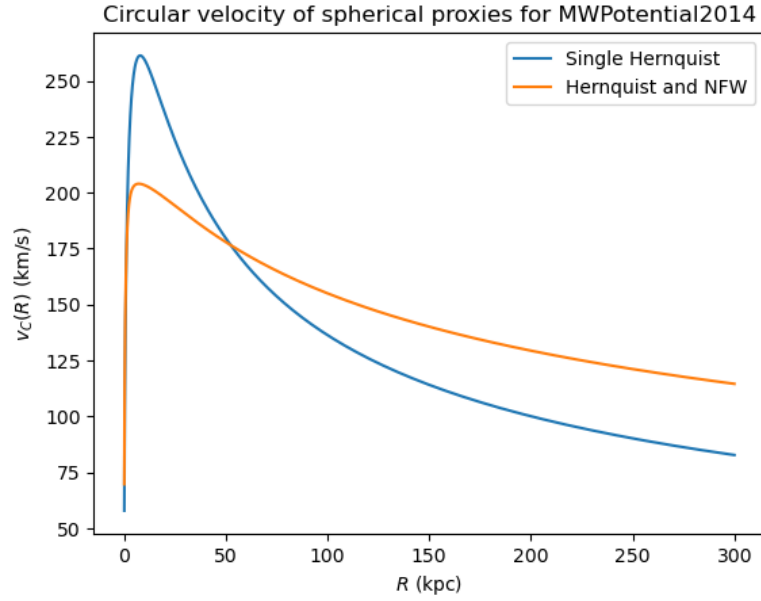


Figure 3.3: Exercise 3, question 3: $v_c(R)$ for single Hernquist and two-component fits for the Milky Way

3.4

For Exercise 3.4, I used `interpSphericalPotential` to build a spherical Milky Way proxy that interpolates MWPotential2014's $M_{enc}(r)$ directly. I also ensured that the percent error in $v_c(R)$ remained below 1% for the entire range of R values (Figure 3.4).

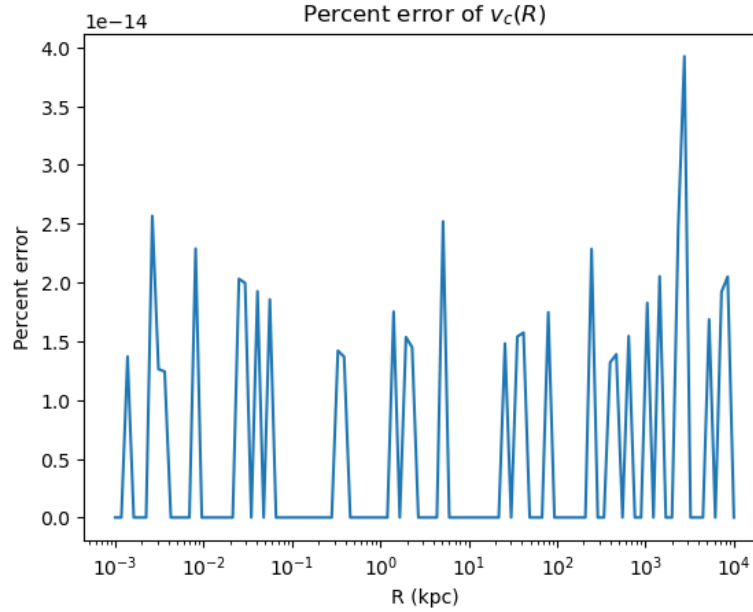


Figure 3.4: Exercise 3, question 4: Percent error in $v_c(R)$ for spherical Milky Way proxy

4 Exercise 4

4.1

For Exercise 4.1, I took my best-fit Hernquist model from Exercise 3 and used it to instantiate the isotropic DF `isotropicHernquistdf`, as well as sampling $N = 10^4$ phase-space samples using the `sample()` method.

4.2

Exercise 4.2 was to check the agreement between the radial number-density profile $n(r)$ and $\rho_{Hern}(r)/m_*$. To get the sample's number-density profile, I separated the potential into spherical shells and got the number of particles in each shell using the `np.histogram` method, and then calculated to volume of each shell. The number-density at each point r could then be calculated

as the number of particles divided by the volume of the shell.

To get the analytical Hernquist number density, I used the formula $\rho_{Hern}(r) = \frac{M}{2\pi} \frac{a}{r(r+a)^3}$, and analyzed it along a fine grid of r values.

Figure 4.1 shows the agreement between $n(r)$ and $\rho_{Hern}(r)/m_*$.

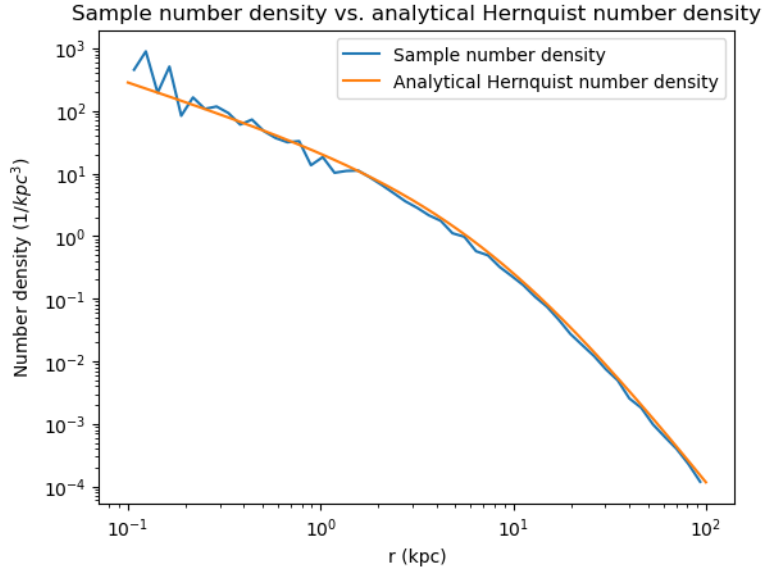


Figure 4.1: Exercise 4, question 2: Radial number density profile for sample vs. analytic Hernquist

4.3

Exercise 4.3 is to compare the velocity-dispersion profile $\sigma_r(r)$ of the sample and the analytic Hernquist prediction (Figure 4.2). For the sample version, I created a for loop to find the inner and outer boundary of each shell, and then make sure that each sampled particle was within that shell, before calculating each velocity-dispersion and saving it to an array. For the analytic version, I used `.sigmar(r)`.

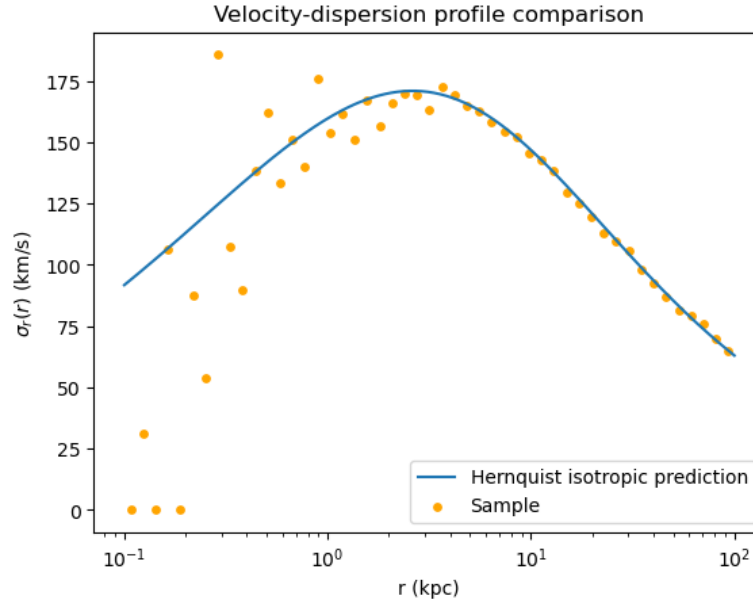


Figure 4.2: Exercise 4, question 3: Comparison between sample velocity dispersion profile and analytic Hernquist isotropic prediction

The sample velocity dispersion matches the Hernquist prediction very well at larger r , although it lacks accuracy at small r ($< 1 \text{ kpc}$).

4.4

For Exercise 4.4, I plotted the velocity-dispersion profiles using constantHernquistBeta for $\beta = +0.3$ and $\beta = -0.3$ to determine how the velocity-dispersion profile would shift and how each β value would affect the radial-to-tangential dispersion ratio.

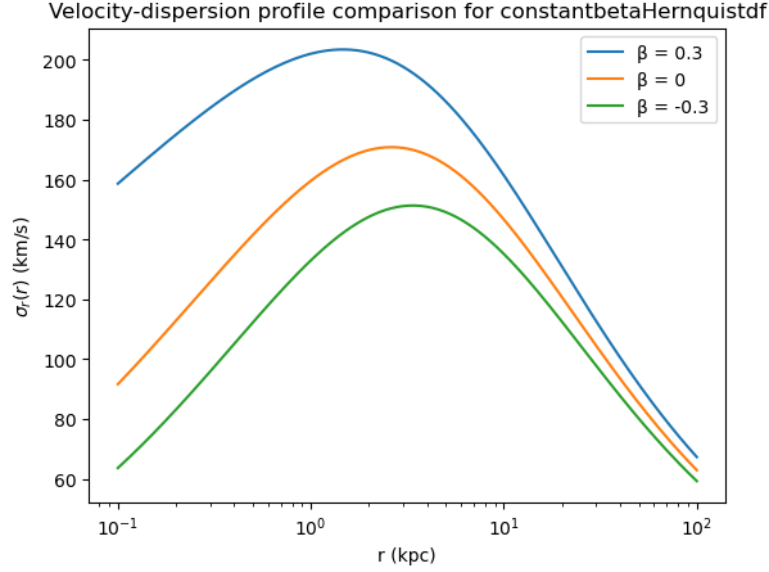


Figure 4.3: Exercise 4, question 4: Change in velocity dispersion profile for $\beta = -0.3$ and $\beta = +0.3$

The positive β value increased the velocity dispersion profile, which signifies more radial dispersion, while the negative β value decreased the velocity dispersion profile, signifying more tangential dispersion.

5 Exercise 5

5.1

For Exercise 5.1, I took a sample of 2000 halo stars and integrated them in my spherical MW potential for 4 Gyr using the C-compiled dop853_c integrator. I used `np.random.seed(0)` throughout the rest of the exercises to maintain a consistent sample both for comparisons and to keep consistent plots when re-running the notebook.

5.2

For Exercise 5.2, I tested the speed-up between using all available cores to parallelize the C integrations vs. using only a single thread.

On average, using all available cores was around 3.5 times faster than using a single thread.

5.3

The purpose of Exercise 5.3 was to plot the initial ($t=0$) and final ($t=4\text{Gyr}$) (x, y) projections of the star sample, as shown in Figures 5.1 and 5.2. Additionally, I overplotted the initial and final radial densities to confirm that the system remained statistically in equilibrium, as shown in Figure 5.3

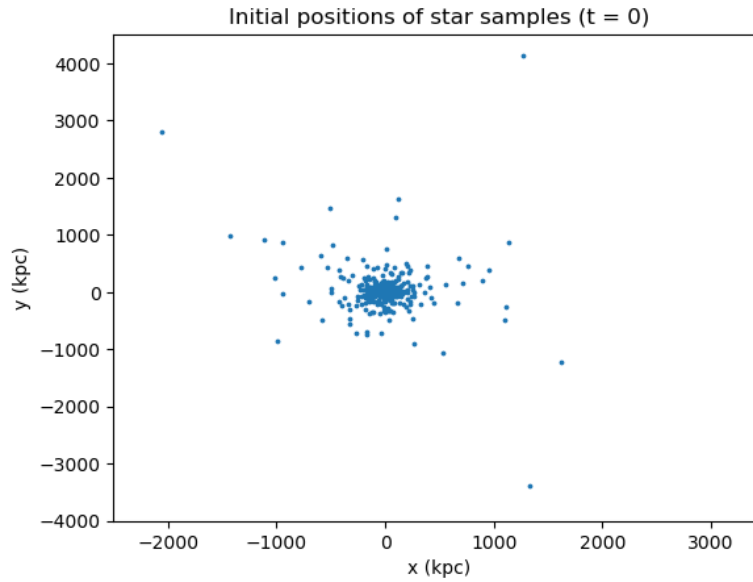


Figure 5.1: Exercise 5, question 3: Initial (x, y) projection of $N=2000$ star sample from isotropic Hernquist DF

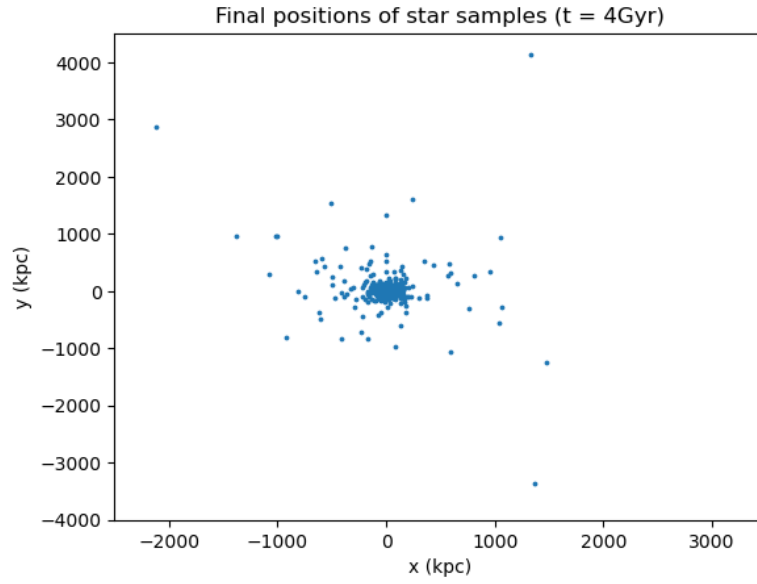


Figure 5.2: Exercise 5, question 3: Final (x, y) projection of $N=2000$ star sample from isotropic Hernquist DF

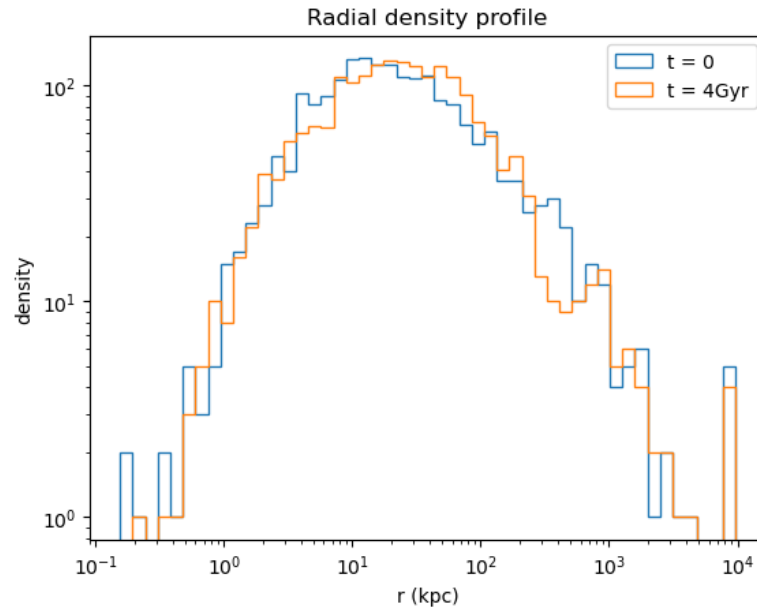


Figure 5.3: Exercise 5, question 3: Radial density profile at $t = 0$ and $t = 4$ Gyr

6 Exercise 6

6.1

Exercise 6.1 was to represent the LMC as a HernquistPotential with total mass $1.5 * 10^{11} M_{Sun}$ and scale radius $a = 10$ kpc, and then to initialize an LMC orbit using its present-day coordinates, which I did using `Orbit.from_name("LMC")`.

6.2

For Exercise 6.2, I integrated the LMC orbit 10 Gyr back in time, including dynamical friction using `ChandrasekharDynamicalFrictionForce`. I then plotted the LMC's galactocentric distance against time, as shown in Figure 6.1.

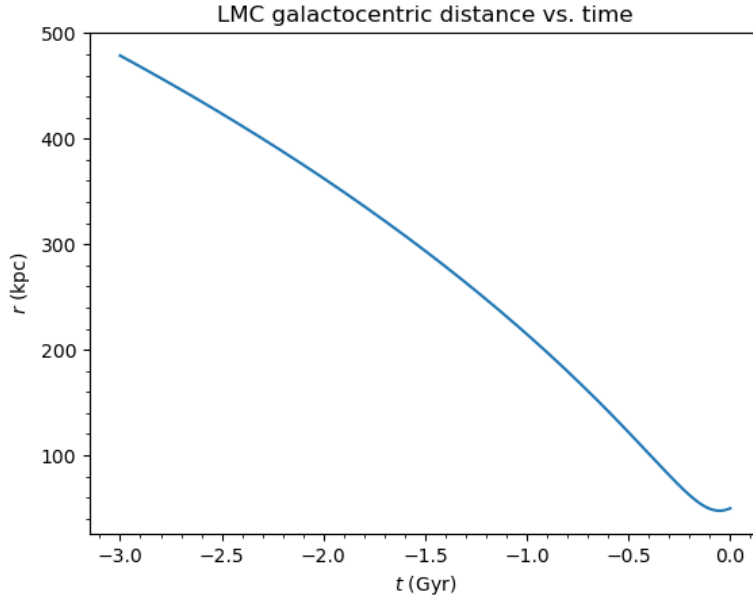


Figure 6.1: Exercise 6, question 2: Galactrocentric distance $r_{LMC}(t)$

6.3

Exercise 6.3 was to integrate the LMC orbit back into the present day using a MovingObjectPotential and verify that its current phase-space coordinates were recovered.

6.4

For Exercise 6.4, I plotted the same $N=200$ halo star sample's (x, y) projection, both with and without the moving LMC projection, shown in Figures 6.2 and 6.3.

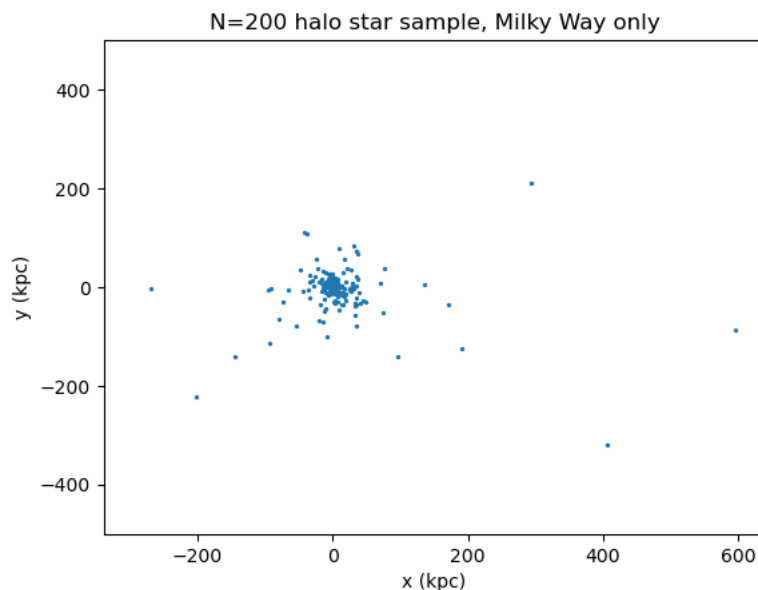


Figure 6.2: Exercise 6, question 4: Unperturbed $N=200$ halo star sample (Milky Way only)

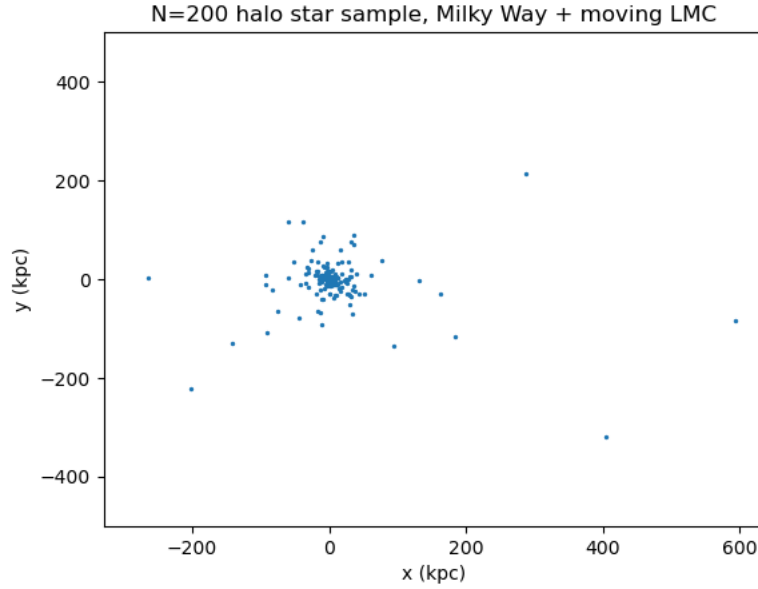


Figure 6.3: Exercise 6, question 4: N=200 halo star sample with Milky Way + moving LMC

Figure 6.3 shows that the halo's outer stars (larger r) were the most affected by the moving LMC perturbation.

7 Exercise 7

Exercise 7 focuses on taking into account the acceleration on the Milky Way caused by the mass of the LMC, which makes the frame centred on the Milky Way non-inertial.

7.1

For Exercise 7.1, I tabulated the acceleration $a_{MW}(t)$'s x , y , and z , components, and plotted them against time. Its magnitude peaks at the pericenter where the LMC is closest to the centre of the Milky Way.

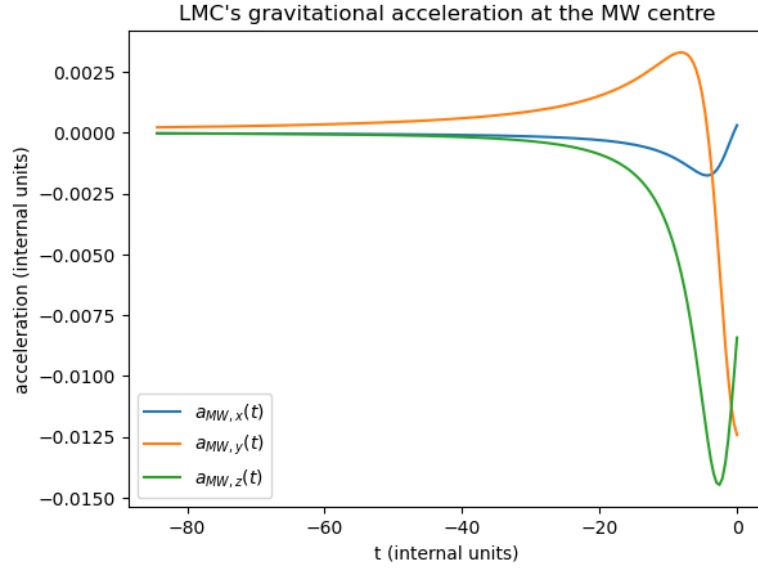


Figure 7.1: Exercise 7, question 1: Acceleration of the Milky Way due to the LMC vs. time

7.2

For Exercise 7.2, I used the tabulated acceleration values to build a `NonInertialFrameForce`. I first built interpolated versions of the accelerations to speed up before building the `NonInertialFrameForce`.

7.3

Exercise 7.3's goal was to rerun the $N=200$ halo integration with the `NonInertialFrameForce` included (Figure 7.2), and see how the dipole pattern changed in comparison to the integration without it (Exercise 6.4).

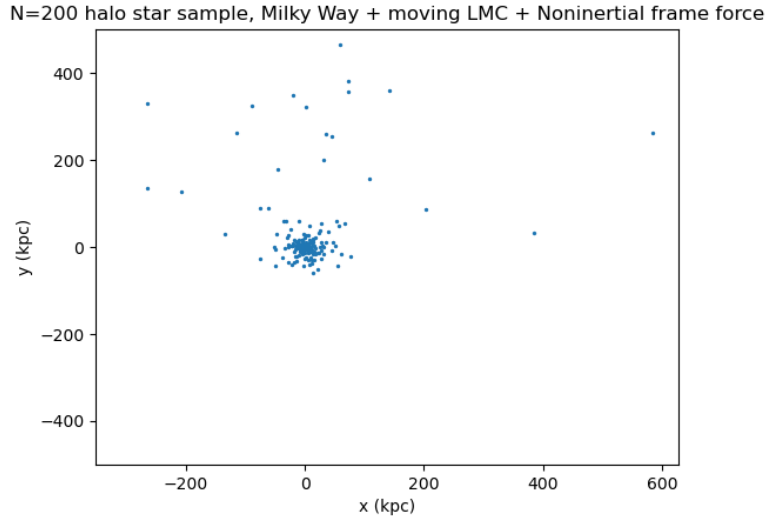


Figure 7.2: Exercise 7, question 3: N=200 Halo integration with Non-inertial frame force included

The non-inertial frame plot (Figure 7.2) shows stars scattered in the $+y$ direction in contrast with the dipole pattern observed in the plot without the non-inertial force included (Figure 6.3)

7.4

Exercise 7.4 asks why the reflex motion of the Milky Way barycenter matters, and how its amplitude depends on the assumed LMC mass.

The LMC's mass pulls the centre of the Milky Way towards it, but the outer halo stars not as strongly bound to the centre and are therefore not accelerated as strongly towards the LMC. This causes the dipole pattern where there is an overdensity of outer halo stars in the $+y$ direction, implying that the LMC is roughly in the $-y$ direction. This effect is missed if the non-inertial frame is not taken into account. Since the LMC's mass is what causes the Milky Way centre to accelerate, the assumed LMC mass is directly related to the magnitude of the acceleration and therefore a more massive LMC increases the amplitude of the reflex term.